NADH dehydrogenase complex functions to accept high energy electrons from NADH. It acts as a proton pump that uses the movement of electrons to move hydrogen ions into the innermembrane space. These clusters are essential for the transfer of hydrogen to the next segment of the chain, Coenzyme Q.

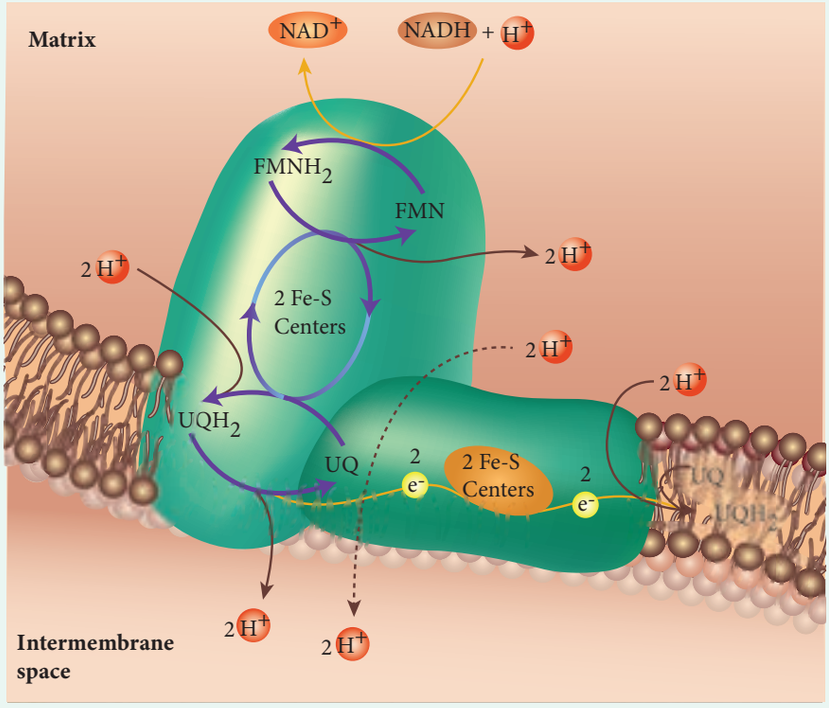


Figure 8.4 NADH dehydrogenase complex

Succinate dehydrogenase complex : This complex II contains the succinate dehydrogenase enzyme used in TCA cycle to transform succinate to fumarate and in the process form $FADH_2$. The $FADH_2$ stays in the complex, and gives two electrons, to a series of Fe-S cluster, that is transferred to ubiquionone (Fig.8.5). This complex is not a proton pump, it does not move the hydrogen ions across the membrane from matrix side to the intermembrane side of mitochondria.

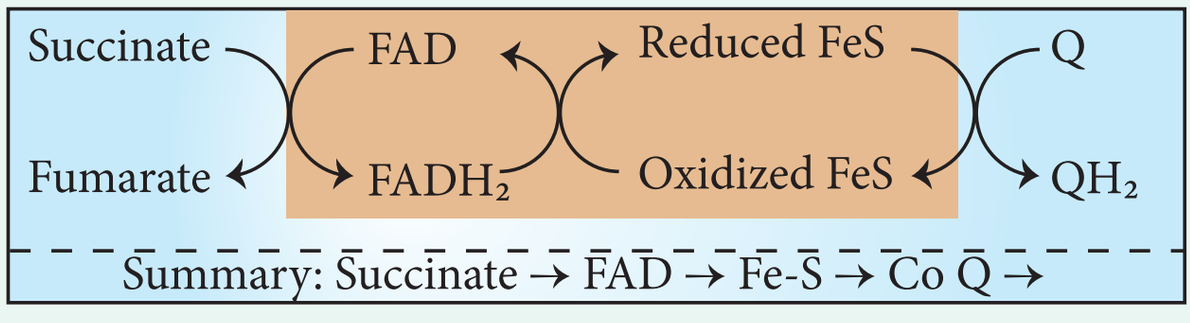


Figure 8.5 Reactions involved in complex II

Coenzyme Q : The succinate dehydrogenase complex of Complex II consists of the TCA cycle enzyme succinate dehydrogenase and two iron sulphur centers. The transfer of electrons from succinate to Coenzyme Q (UQ) is mediated via the iron sulphur centers. Co Q can accept hydrogen atoms from $FMNH_2$ of NADH dehydrogenase (Complex I) and $FADH_2$ from succinate dehydrogenase (Complex II). Further, electrons from cytoplasmic NADH are also transferred to ubiquinone (UQ) through glycerol 3 phosphate dehydrogenase and acyl coA dehydrogenase (Fig.8.6).

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